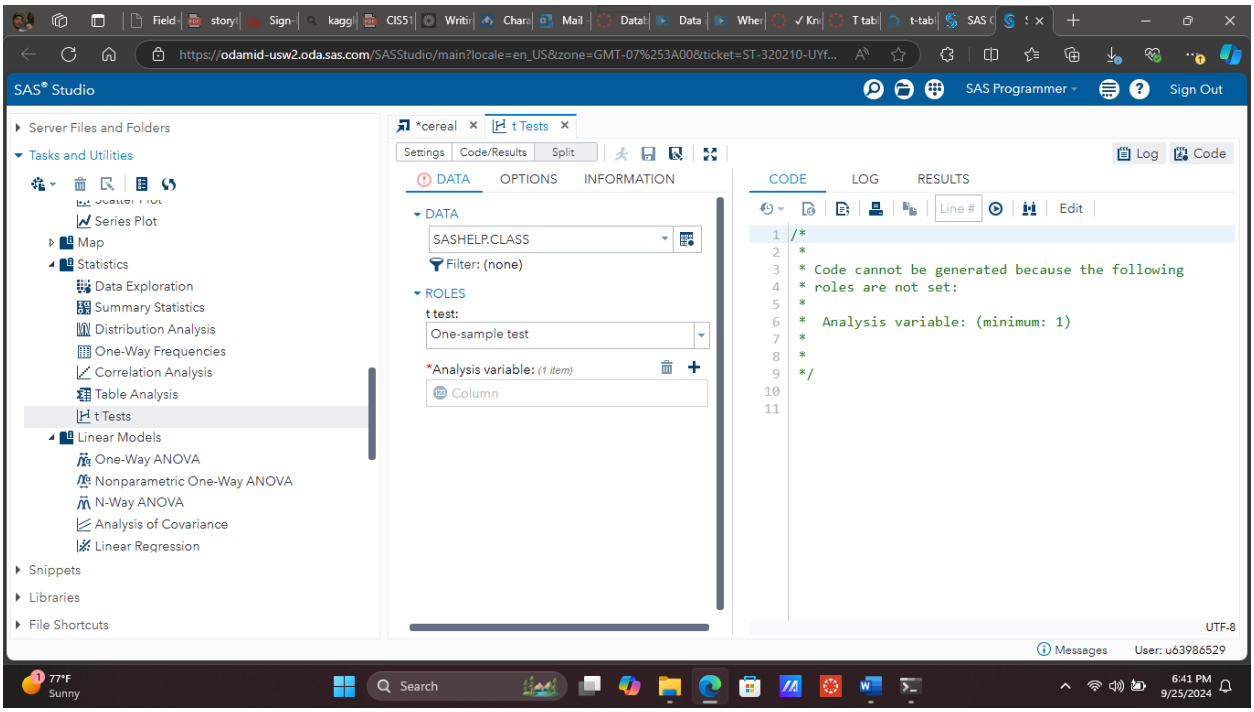
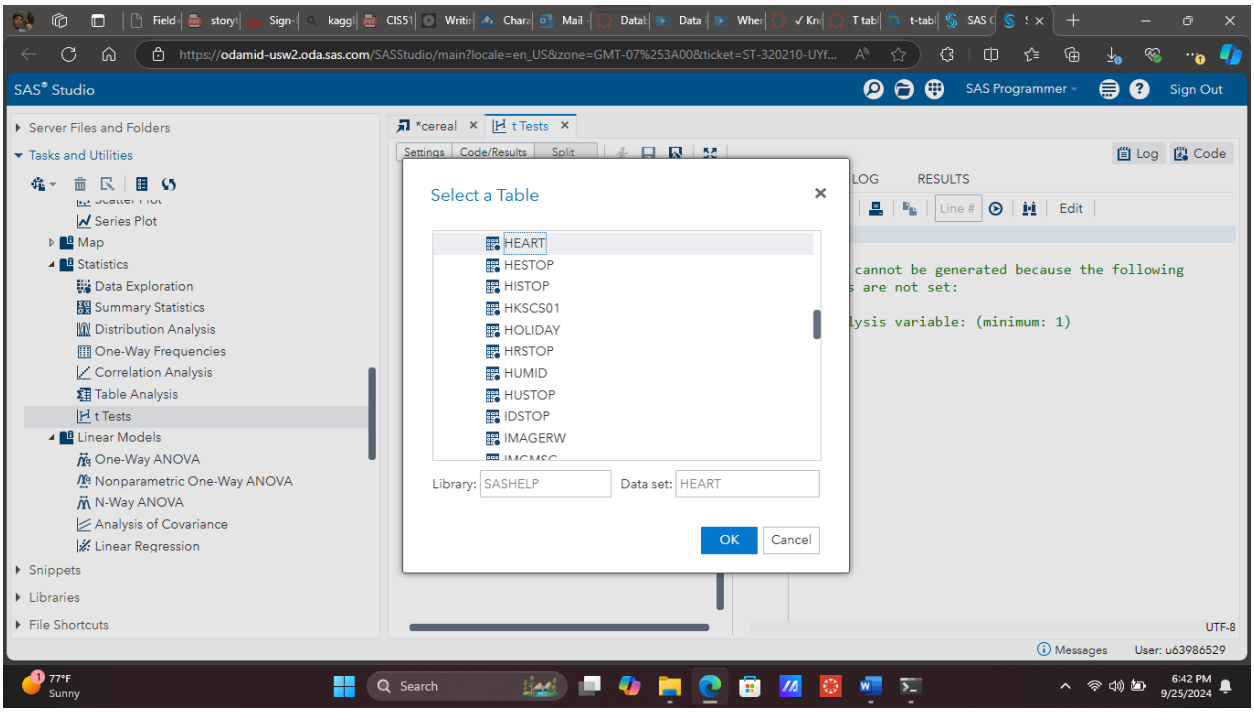
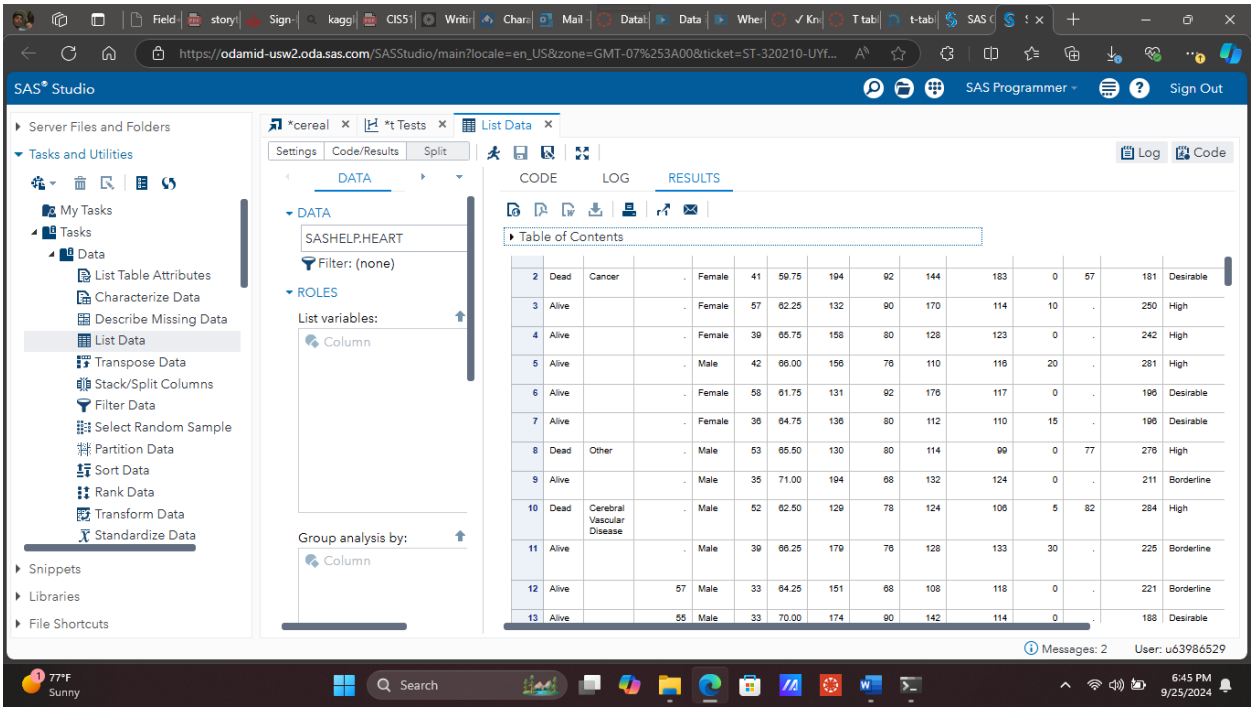
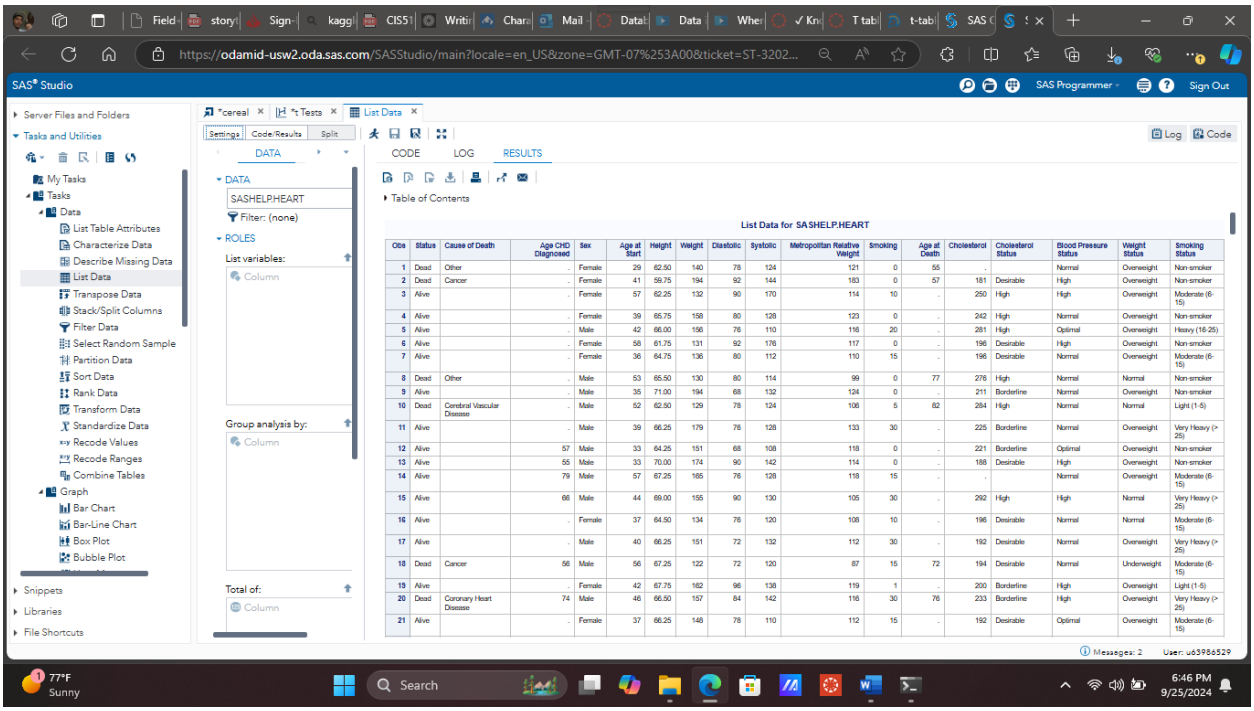
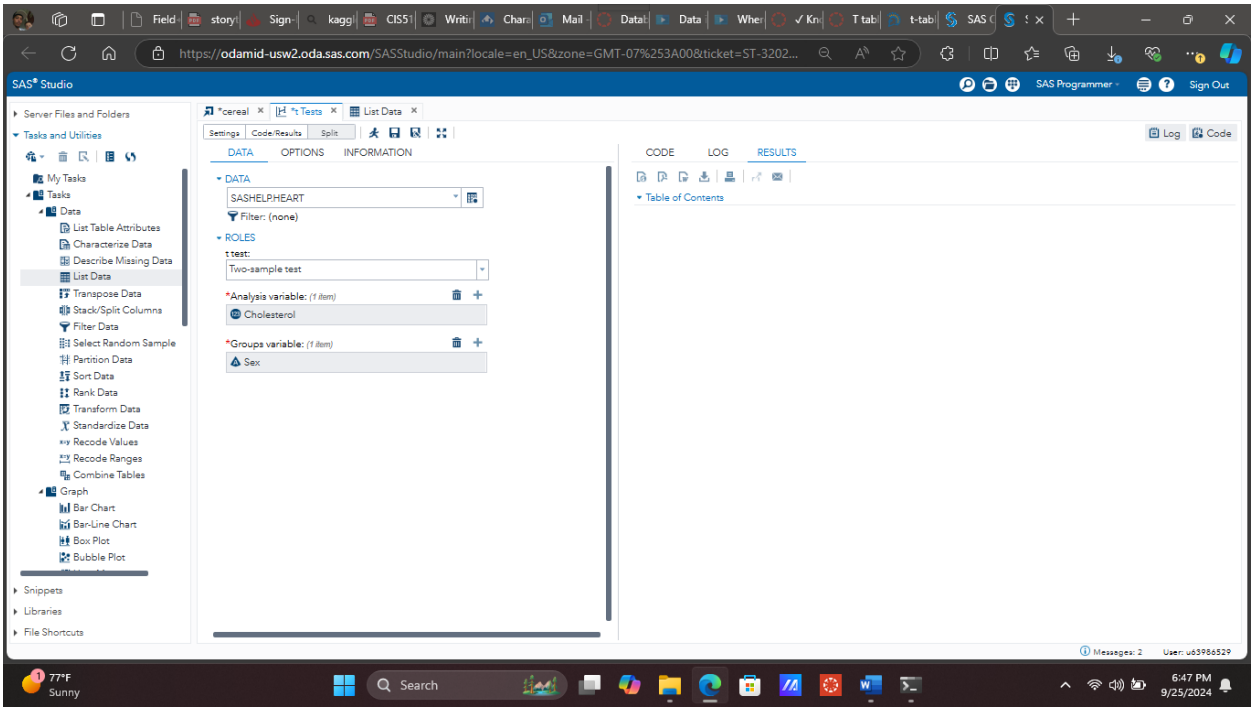
Lab 8

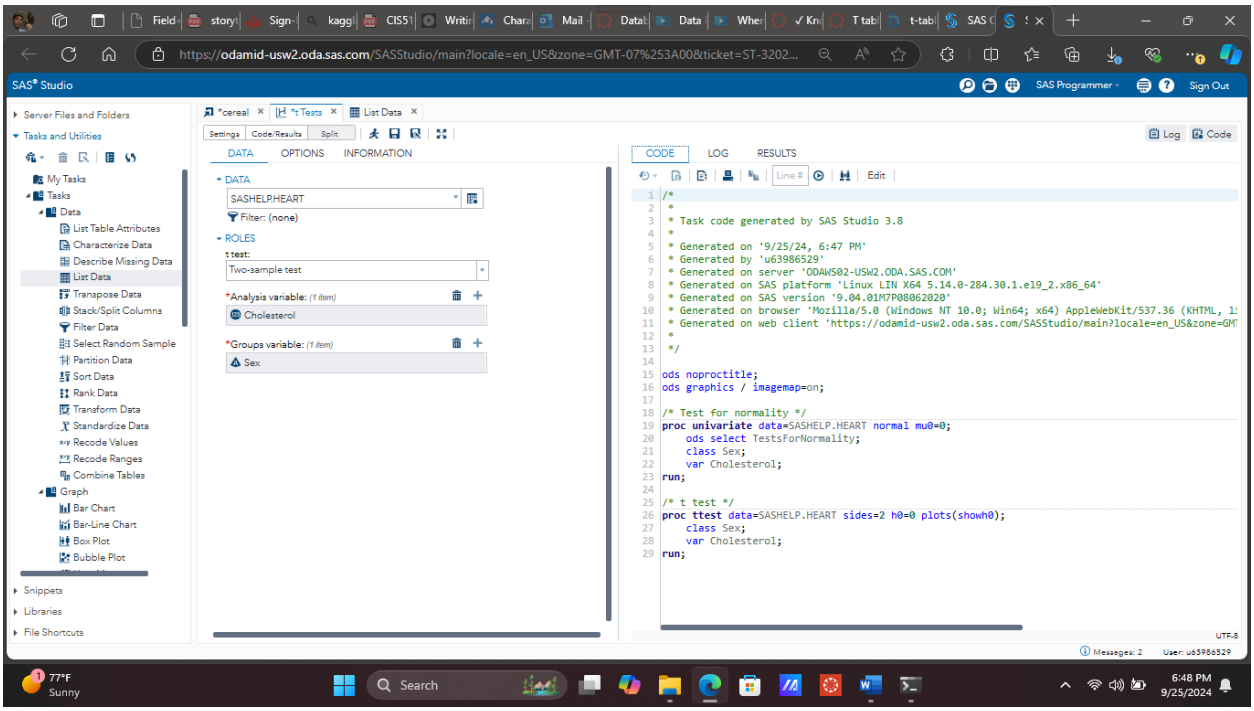

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One-Way ANOVA
Nonparametric One-Way ANOVA
N-Way ANOVA
Analysis of Covariance
Linear Regression
Snippets
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*cereal
t Tests
Settings
Code/Results
Split
DATA
OPTIONS
INFORMATION
SASHELP.CLASS
Filter: (none)
ROLES
t test:
One-sample test
*Analysis variable: (1 item)
Column
Log
Code
CODE
LOG
RESULTS
Line #
Edit
/*
* Code cannot be generated because the following
* roles are not set:
* Analysis variable: (minimum: 1)
*/
UTF-8
Messages
User: u63986529
77°F
Sunny
Search
6:41 PM
9/25/2024

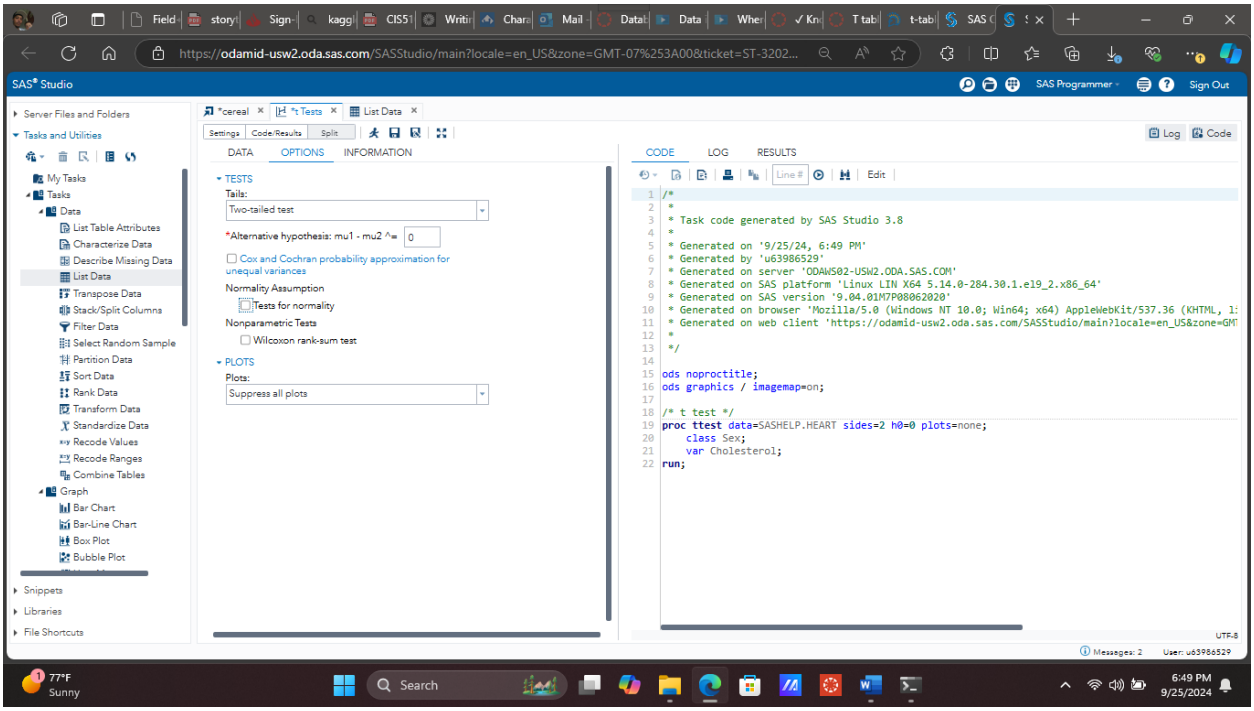

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t Tests
Linear Models
One-Way ANOVA
Nonparametric One-Way ANOVA
N-Way ANOVA
Analysis of Covariance
Linear Regression
Snippets
Libraries
File Shortcuts
*cereal
t Tests
Log
Code
RESULTS
Edit
Select a Table
HEART
HESTOP
HISTOP
HKSCS01
HOLIDAY
HRSTOP
HUMID
HUSTOP
IDSTOP
IMAGERW
Library: SASHELP
Data set: HEART
OK
Cancel
cannot be generated because the following
are not set:
lysis variable: (minimum: 1)
UTF-8
Messages
User: u63986529
77°F
Sunny
Search
6:42 PM
9/25/2024

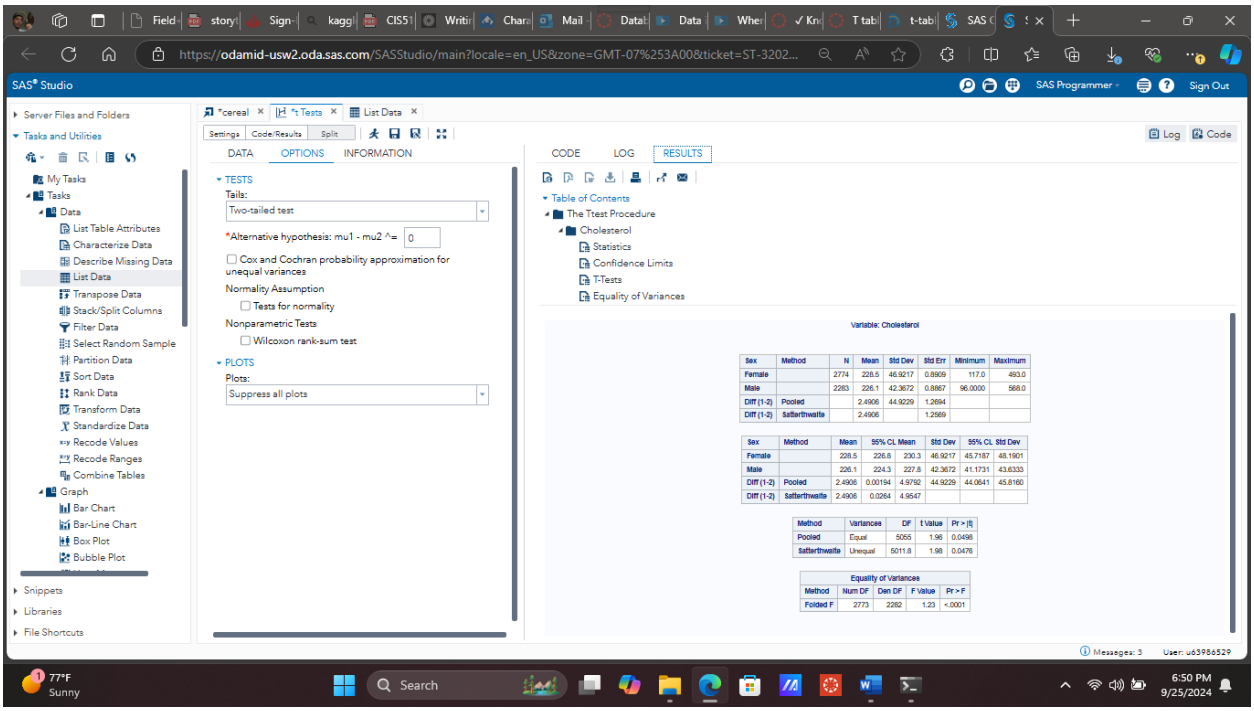

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Rank Data
Transform Data
Standardize Data
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*cereal
*t Tests
List Data
Settings
Code/Results
Split
Log
Code
DATA
CODE
LOG
RESULTS
SASHELP.HEART
Filter: (none)
ROLES
List variables:
Column
Group analysis by:
Table of Contents
Messages: 2
User: u63986529

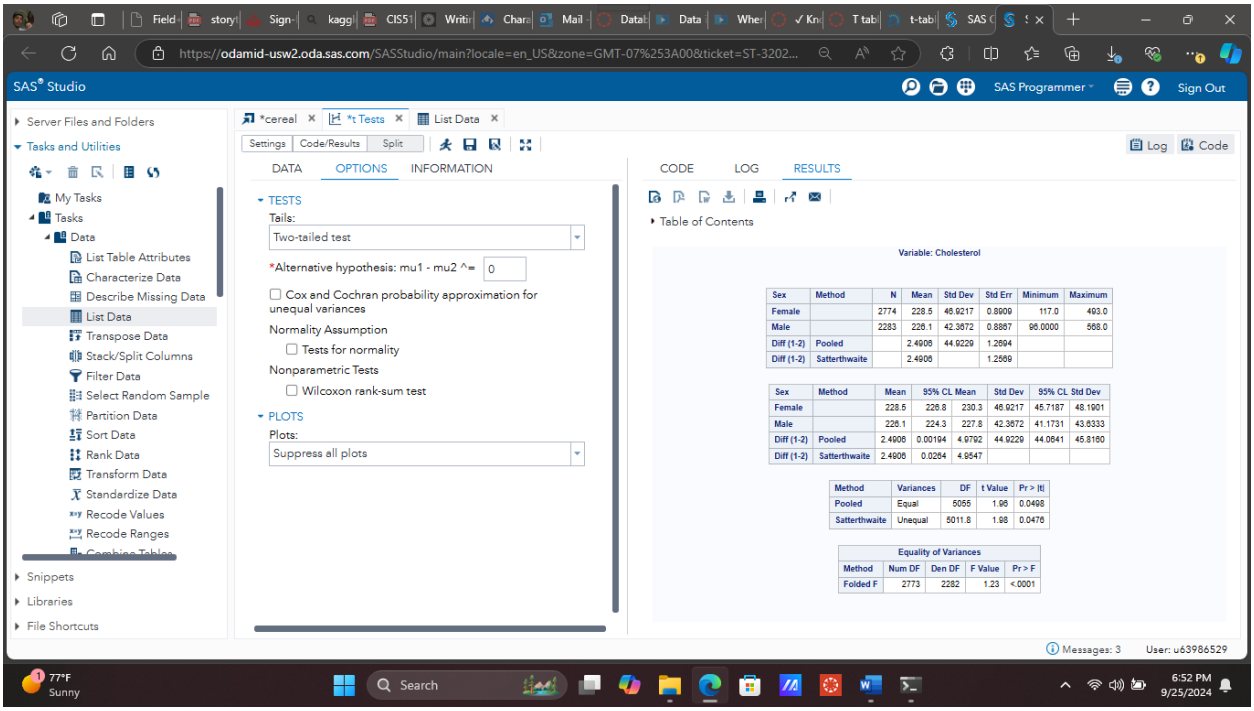

SAS® Studio
SAS Programmer
Sign Out
List Data for SASHELP.HEART
Obs
Status
Cause of Death
Age CHD Diagnosed
Sex
Age at Start
Height
Weight
Diastolic
Systolic
Metropolitan Relative Weight
Smoking
Age at Death
Cholesterol
Cholesterol Status
Blood Pressure Status
Weight Status
Smoking Status
Total of:
Messages: 2
User: u63986529


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Standardize Data
Recode Values
Recode Ranges
Combine Tables
Graph
Bar Chart
Bar-Line Chart
Box Plot
Bubble Plot
Snippets
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*cereal
*t Tests
List Data
Settings
Code/Results
Split
DATA
OPTIONS
INFORMATION
DATA
SASHELP.HEART
Filter: (none)
ROLES
t test:
Two-sample test
*Analysis variable: (1 item)
Cholesterol
*Groups variable: (1 item)
Sex
Log
Code
CODE
LOG
RESULTS
Table of Contents
77°F
Sunny
Search
6:47 PM
9/25/2024


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Transform Data
Standardize Data
Recode Values
Recode Ranges
Combine Tables
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Bubble Plot
Snippets
Libraries
File Shortcuts
*cereal
*t Tests
List Data
DATA
OPTIONS
INFORMATION
SASHELP.HEART
Filter: (none)
Two-sample test
Cholesterol
Sex
Log
Code
CODE
LOG
RESULTS
/*
* Task code generated by SAS Studio 3.8
* Generated on '9/25/24, 6:47 PM'
* Generated by 'u63986529'
* Generated on server 'ODAWS02-USW2.ODA.SAS.COM'
* Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64'
* Generated on SAS version '9.04.01M7P08062020'
* Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, 1
* Generated on web client 'https://odamid-usw2.oda.sas.com/SASStudio/main?locale=en_US&zone=GM
*
*/
ods noproctitle;
ods graphics / imagemap=on;
/* Test for normality */
proc univariate data=SASHELP.HEART normal mu0=0;
ods select TestsForNormality;
class Sex;
var Cholesterol;
run;
/* t test */
proc ttest data=SASHELP.HEART sides=2 h0=0 plots(showh0);
class Sex;
var Cholesterol;
run;
77°F
Sunny
Search
6:48 PM
9/25/2024


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Recode Values
Recode Ranges
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*cereal
*t Tests
List Data
Settings
Code/Results
Split
DATA
OPTIONS
INFORMATION
TESTS
Tails:
Two-tailed test
*Alternative hypothesis: mu1 - mu2 ^= 0
Cox and Cochran probability approximation for unequal variances
Normality Assumption
Tests for normality
Nonparametric Tests
Wilcoxon rank-sum test
PLOTS
Plots:
Suppress all plots
Log
Code
CODE
LOG
RESULTS
Edit
/*
* Task code generated by SAS Studio 3.8
*
* Generated on '9/25/24, 6:49 PM'
* Generated by 'u63986529'
* Generated on server 'ODAWS02-USW2.ODA.SAS.COM'
* Generated on SAS platform 'Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64'
* Generated on SAS version '9.04.01M7P08062020'
* Generated on browser 'Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, l
* Generated on web client 'https://odamid-usw2.oda.sas.com/SASStudio/main?locale=en_US&zone=GM
*
*/
ods noproctitle;
ods graphics / imagemap=on;
/* t test */
proc ttest data=SASHELP.HEART sides=2 h0=0 plots=none;
class Sex;
var Cholesterol;
run;
UTF-8
Messages: 2
User: u63986529
77°F
Sunny
Search
6:49 PM
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Tails:
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*Alternative hypothesis: mu1 - mu2 ^= 0
Cox and Cochran probability approximation for unequal variances
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PLOTS
Plots:
Suppress all plots
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Code
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Table of Contents
The Ttest Procedure
Cholesterol
Statistics
Confidence Limits
T-Tests
Equality of Variances
Variable: Cholesterol
Sex Method N Mean Std Dev Std Err Minimum Maximum
Female 2774 228.5 46.9217 0.8909 117.0 493.0
Male 2283 226.1 42.3672 0.8867 96.0000 568.0
Diff (1-2) Pooled 2.4906 44.9229 1.2694
Diff (1-2) Satterthwaite 2.4906 1.2569
Sex Method Mean 95% CL Mean Std Dev 95% CL Std Dev
Female 228.5 226.8 230.3 46.9217 45.7187 48.1901
Male 226.1 224.3 227.8 42.3672 41.1731 43.6333
Diff (1-2) Pooled 2.4906 0.00194 4.9792 44.9229 44.0641 45.8160
Diff (1-2) Satterthwaite 2.4906 0.0284 4.9547
Method Variances DF t Value Pr > |t|
Pooled Equal 5055 1.96 0.0498
Satterthwaite Unequal 5011.8 1.98 0.0478
Equality of Variances
Method Num DF Den DF F Value Pr > F
Folded F 2773 2282 1.23 <.0001
Messages: 3
User: u63986529
77°F
Sunny
Search
6:50 PM
9/25/2024



Answer:

Ho: There is no difference in cholesterol level between the genders.

Ha: There is a difference in cholesterol level between the genders.

0.0476 < 0.05, REJECT Ho

There is a significant difference in cholesterol level between the genders.